



TELECOM CUSTOMER CHURN PREDICTION

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INTRODUCTION

- **Customer Churn** refers to the phenomenon where **customers discontinue** or **cancel** their **subscription** to a service.
- In the telecommunications industry, churn is a **critical challenge** due to **intense market competition** and availability of multiple service providers.
- **Retaining existing customers is more cost-effective** than acquiring new ones, making churn prediction highly valuable.
- With the growth of big data and data mining techniques, companies can **analyze customer behavior** and **identify hidden patterns**.
- Predictive analytics and machine learning models enable organizations to **forecast churn probability** in advance.
- This project focuses on using historical customer data to build an intelligent model that helps in **early detection of churn and supports data-driven decision making**.

OBJECTIVE OF THE PROJECT

To predict whether a customer will churn or not

To identify key factors influencing churn.

To help telecom companies improve customer retention

To build a machine learning model for classification.



Dataset OVERVIEW

Source

KAGGLE

Records

10000+

Features

20+

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
1	customer	data_known	gender	age	tariff	dropped_calls	handset	peak_mins	peak_rate	bill_peak	offpeak_mi	offpeak_rate	bill_offpeak	gadget_A	gadget_B	gadget_C	gadget_D	gadget_A_revenue	gadget_B_revenue	gadget_C_revenue	gadget_D_revenue	
2	N247880	yes	Female	31	CAT 50	0	BS110	393.6	15	59.04	19.2	5	0.96	F	F	F	F	0	0	0	0	
3	N376450	yes	Female	42	CAT 50	6	CAS30	136.2	15	20.43	36.9	5	1.845	F	F	F	F	0	0	0	0	
4	N109880	yes	Female	62	CAT 100	9	ASAD170	657.6	15	98.64	122.4	5	6.12	T	T	T	F	11	20	28	0	

VARIABLES

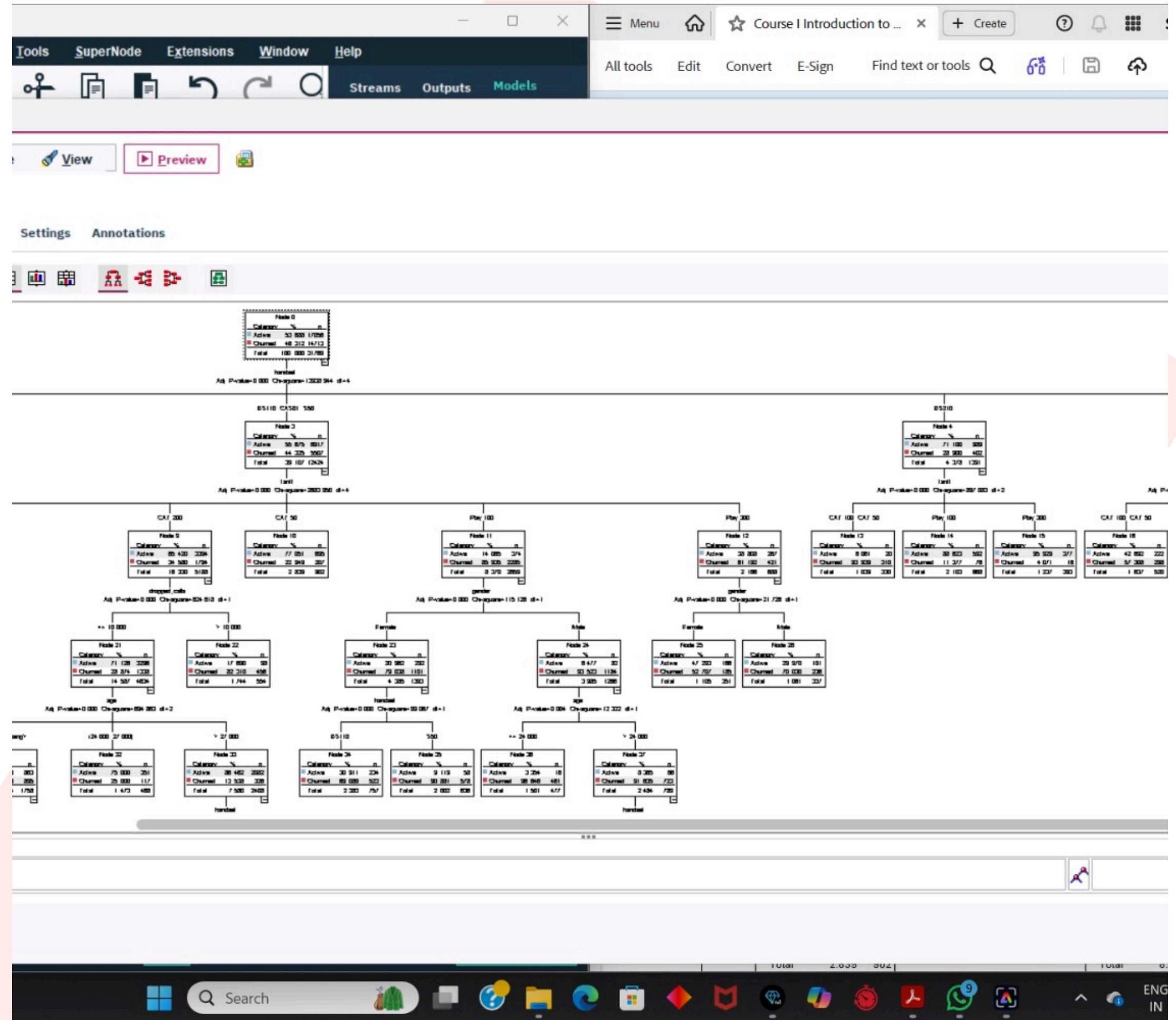
DESCRIPTION

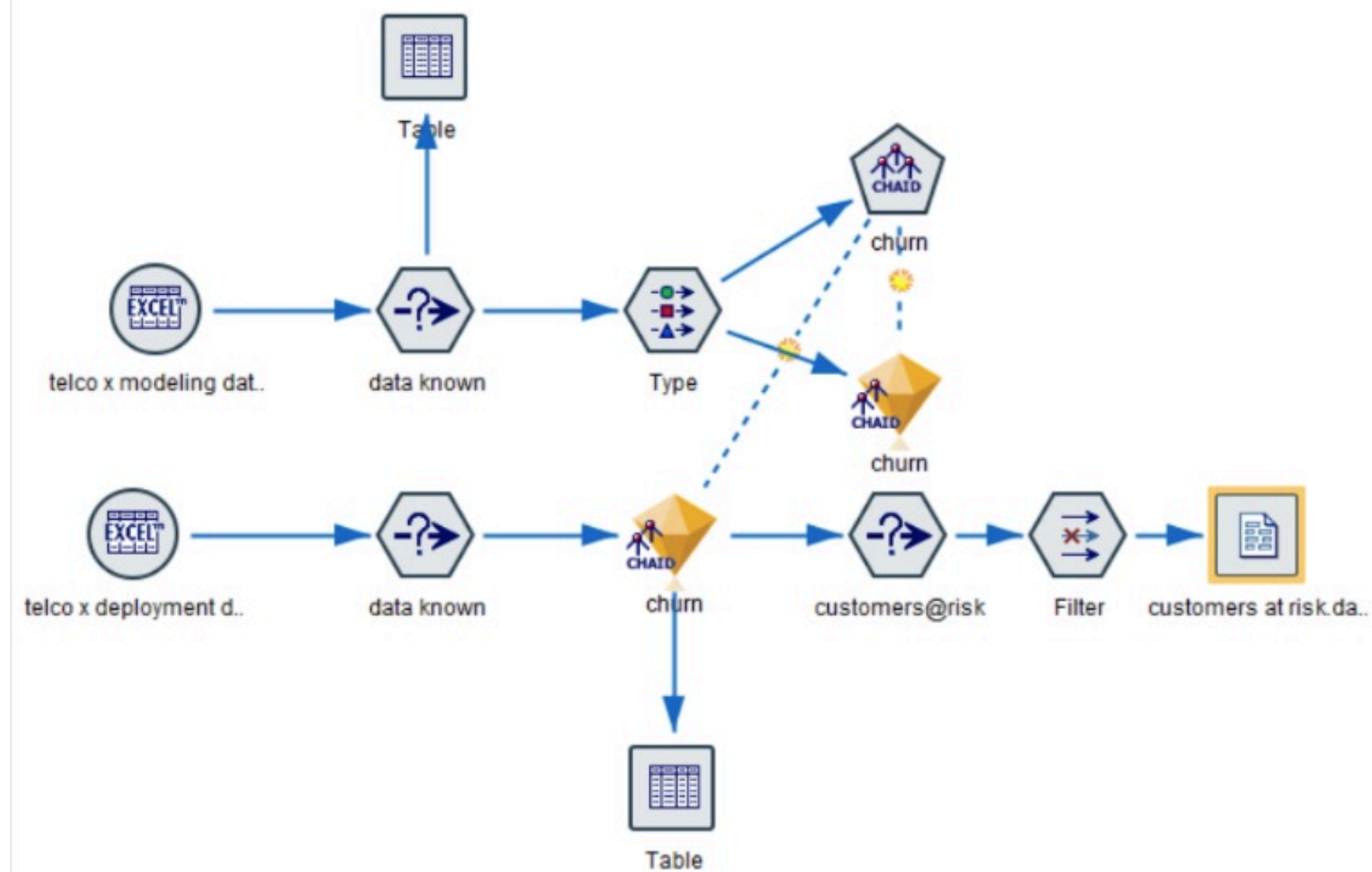
	A	B	C	D	E	F	G	H	I	J	K	L
1	customer_id	data_known	gender	age	tariff	dropped_calls	handset	peak_mins	peak_rate	bill_peak	offpeak_mi	offpeak_rate
2	N247880	yes	Female	31	CAT 50	0	BS110	393.6	15	59.04	19.2	5
3	N376450	yes	Female	42	CAT 50	6	CAS30	136.2	15	20.43	36.9	5
4	N109880	yes	Female	62	CAT 100	9	ASAD170	657.6	15	98.64	122.4	5

M	N	O	P	Q	R	S	T	U
bill_offpeak	gadget_A	gadget_B	gadget_C	gadget_D	gadget_A_revenue	gadget_B_revenue	gadget_C_revenue	gadget_D_revenue
0.96	F	F	F	F	0	0	0	0
1.845	F	F	F	F	0	0	0	0
6.12	T	T	T	F	11	20	28	0

MISSION

- The mission of this project is to predict telecom customers who are likely to churn using SPSS predictive analytics techniques.
- It aims to analyze customer usage patterns, service history, and demographic data to identify high-risk customers.
- The project helps telecom companies take preventive actions and improve customer retention strategies.
- Our goal is to reduce customer loss and increase business profitability through data-driven decision making.



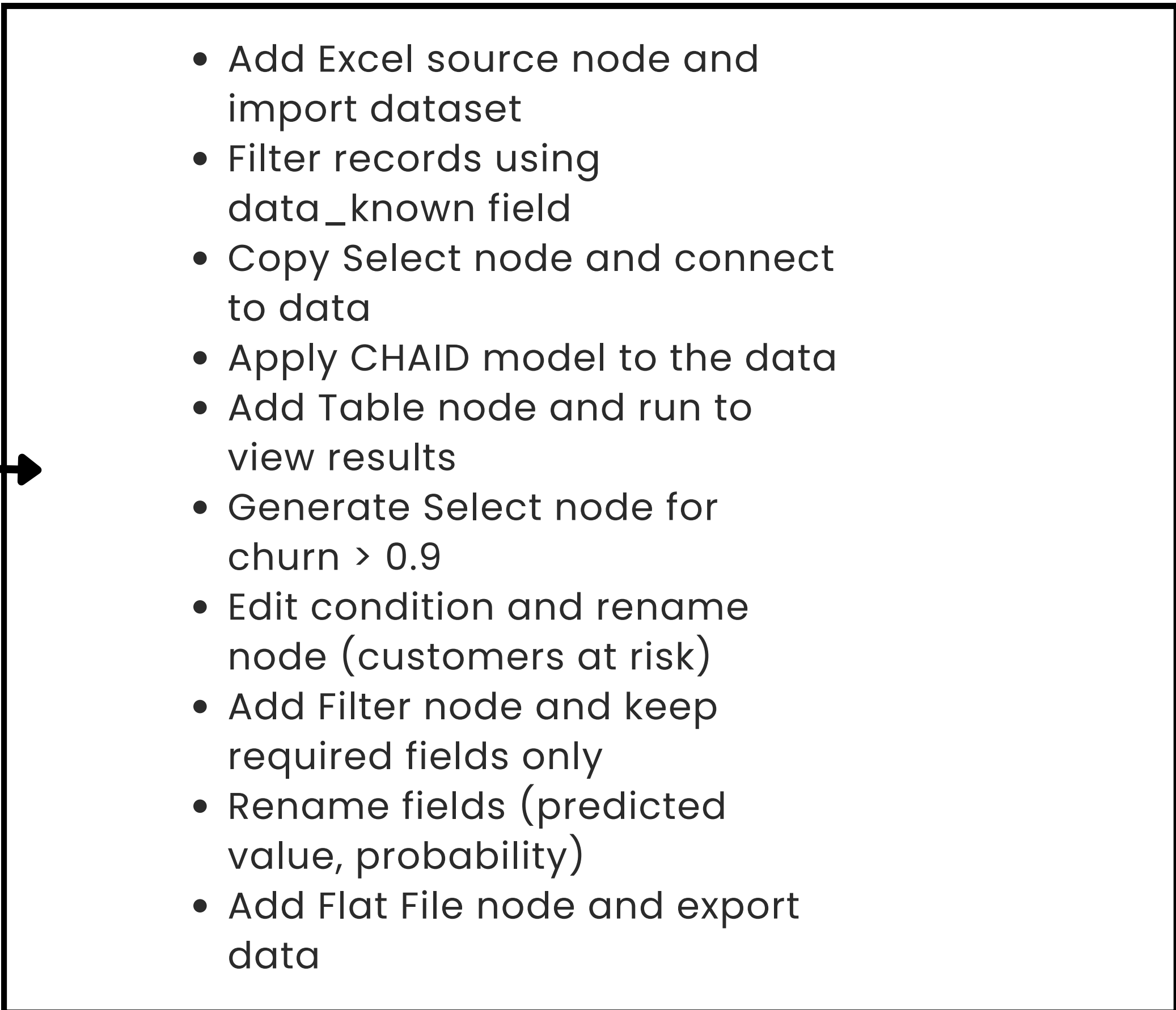


Preview from telco x customer d...
Preview from telco x customer d...
Preview from telco x modeling d...
Table (23 fields, 31,789 records)
Table (23 fields, 31,789 records...
Table (23 fields, 31,789 records...
Table (23 fields, 1 records)
Table (23 fields, 1 records) #1
Preview from data known Node ...
Preview from data known Node ...
Table (23 fields, 1 records) #2
Table (23 fields, 31,789 records...
Table (23 fields, 31,769 records)
Table (23 fields, 31,769 records...
Preview from churn Node (25 fie...
Preview from churn Node (25 fie...
Preview from telco x modeling d...

(unsaved project)
Business Understanding
Data Understanding
Data Preparation
Modeling
Evaluation
Deployment

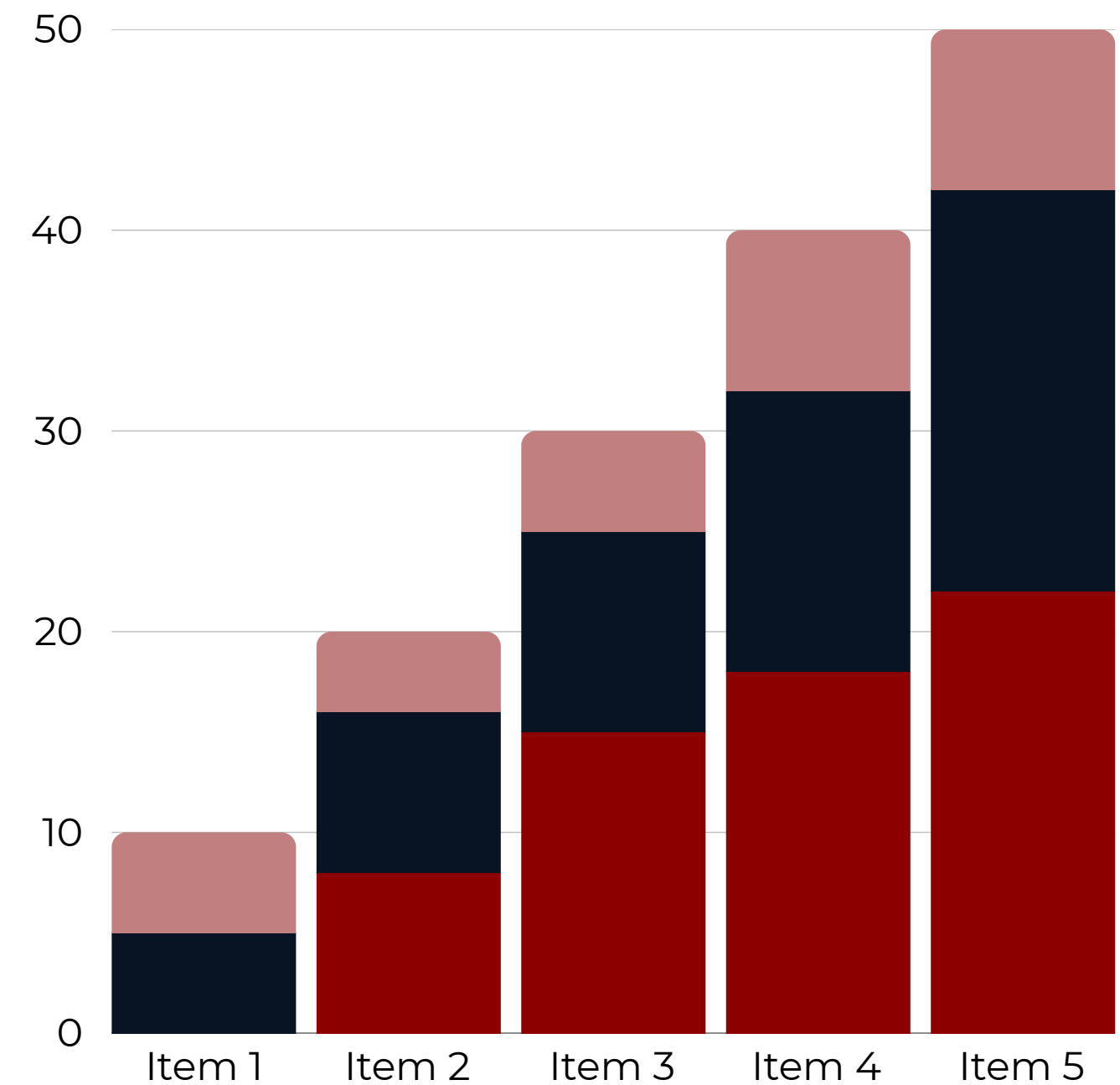
The Process



- Add Excel source node and import dataset
 - Filter records using data_known field
 - Copy Select node and connect to data
 - Apply CHAID model to the data
 - Add Table node and run to view results
 - Generate Select node for churn > 0.9
 - Edit condition and rename node (customers at risk)
 - Add Filter node and keep required fields only
 - Rename fields (predicted value, probability)
 - Add Flat File node and export data
- 

MARKET ANALYSIS

The telecom industry is highly competitive, with customers frequently switching providers for better pricing, network quality, and service benefits. Customer churn directly affects company revenue and increases the cost of acquiring new customers. Analyzing customer behavior, usage patterns, and complaints helps identify the major reasons behind churn. Predictive analytics enables telecom companies to improve retention strategies and maintain revenue growth.



Our **GOALS**

- Identify factors causing customer churn
- Predict customers likely to leave
- Improve customer retention strategies
- Analyze key influencing variables
- Support data-driven decisions
- Segment customers by churn risk
- Evaluate model accuracy



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THANK YOU